

Requirements Engineering Framework

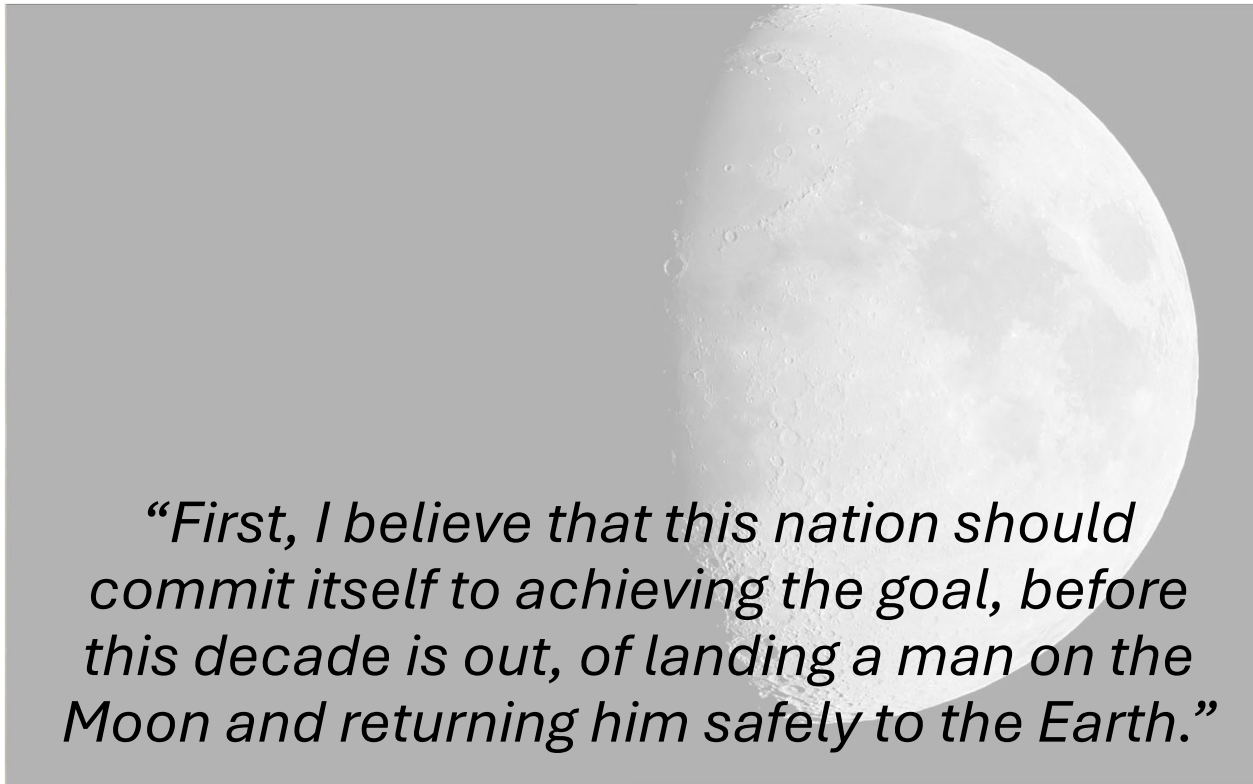
The “Vision”

- Idea/need/wish to change the current reality!



Establishing Vision in Context

The “Vision” - Example



“First, I believe that this nation should commit itself to achieving the goal, before this decade is out, of landing a man on the Moon and returning him safely to the Earth.”

**Vision of
J.F. Kennedy
in 1961**

[Dudley 2000]

Pictures from [2, 3]

Establishing Vision in Context

Example of a (System-)Vision



“First, I believe that this nation should commit itself to achieving the goal, before this decade is out, of landing a man on the Moon ~~and returning him safely to the Earth.~~”

Small changes to the vision can have a big impact on the requirements and the way the system is developed!

Vision of J.F. Kennedy in 1961 [Dudley 2000]

Establishing Vision in Context

Vision

- Describe a clearly defined and verifiable goal
- Defines an intended (small or large) change to a current reality
- Typically brief and precise
- Guides the definition of requirements
- Guides the development of the system
- States a goal (“What?”),
not how to achieve it (“How?”)
- Basis for making decisions
- Justifies expenses

Establishing Vision in Context

Vision Guides Initial Context Consideration



Establishing Vision in Context

Vision Guides Initial Context Consideration



Establishing Vision in Context

Establishing a Vision in Context

- Every system is embedded in a specific context
- The requirements sources to be considered, depend on the context
- The context strongly influences the definition and refinement of requirements
- Adequate consideration of the context is thus essential for requirements engineering and system development
- The context is (typically) not fully known/understood

Example: Power Supply (1)

E **Vision:** Develop a power supply based on renewable energies.

Context 1: Family home in a countryside

→ **Wind turbine** is a **good solution**,
provided there is enough
wind in the area



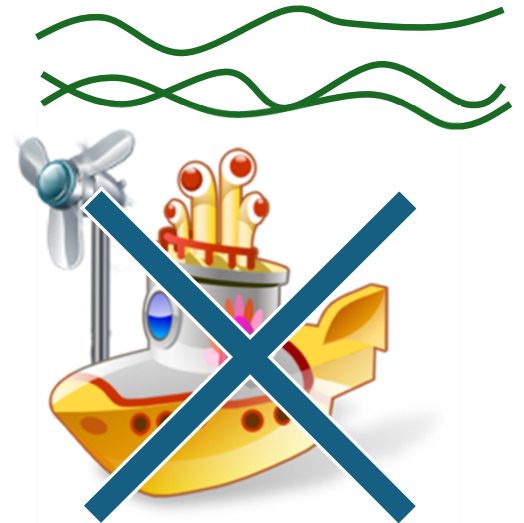
Establishing Vision in Context

Example: Power Supply (2)

E **Vision:** Develop a power supply based on renewable energies.

Context 2: Submarine under water

→ **Wind turbine** is an **infeasible solution**
(cannot be use under water)



All icons from [1]

Establishing Vision in Context

Example: Power Supply (3)

E **Vision:** Develop a power supply based on renewable energies.

Context 3: Satellite in space

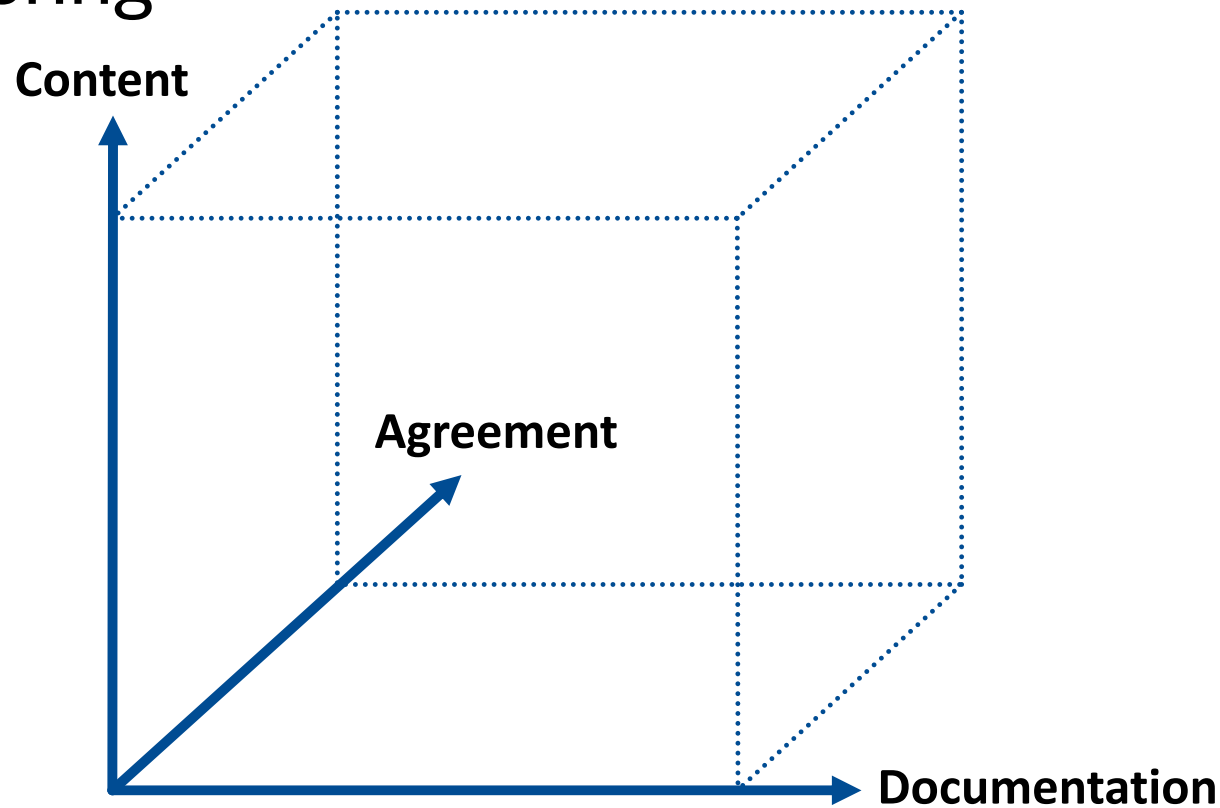
→ **Wind turbine** is
an **infeasible solution**
(there is no wind in space)



All icons from [1]

Goal of Requirements Engineering – Three Dimensions of Requirements Engineering

Three Dimensions of Requirements Engineering



Based on [Pohl and Ulfat-Bunyadi 2013]

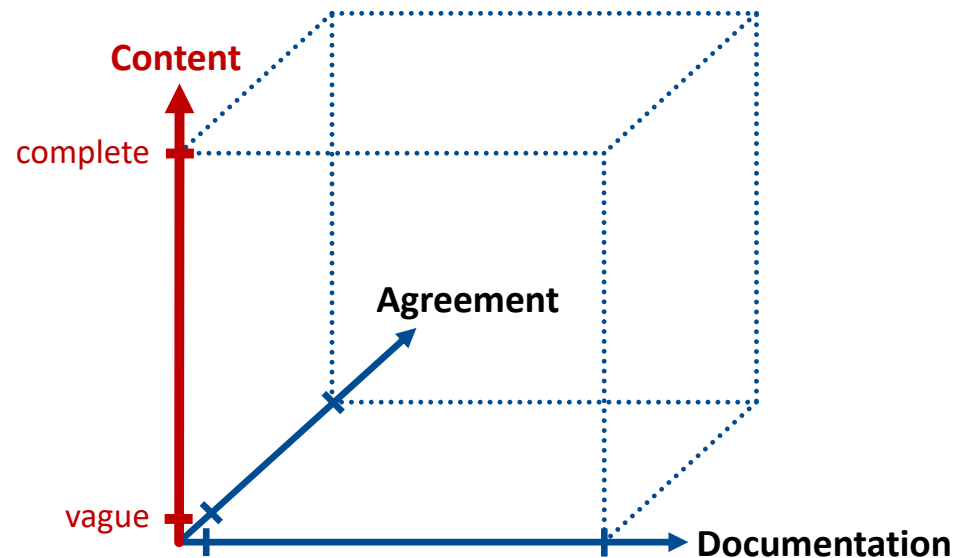
Goal of Requirements Engineering – Three Dimensions of Requirements Engineering

Content Dimension

- Focus: Understanding of the requirements and the context
- At the beginning of the requirements engineering process: Only a few requirements and context information are known; understanding is typically vague



At the end:
All relevant requirements
and context information are
explicitly known and
understood at the
required level of detail



Based on [Pohl and
Ulfat-Bunyadi 2013]

Goal of Requirements Engineering – Three Dimensions of Requirements Engineering

Documentation Dimension

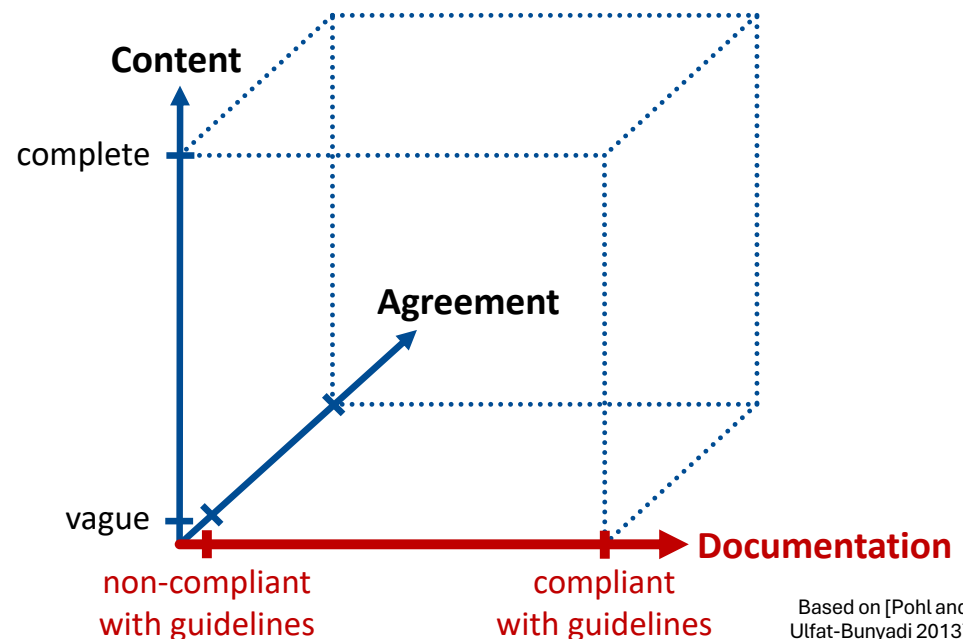
- Focus: Documenting/specifying relevant information (including requirements) using appropriate documentation/specification formats
- At the beginning of the requirements engineering process: Information typically documented non-compliant with documentation/specification guidelines,

e.g., a note, sketch, statement in a minute, hand drawing, ...



At the end:

All requirements and context information should be documented/specified in compliance with the relevant documentation/specification guidelines



Based on [Pohl and Ulfat-Bunyadi 2013]

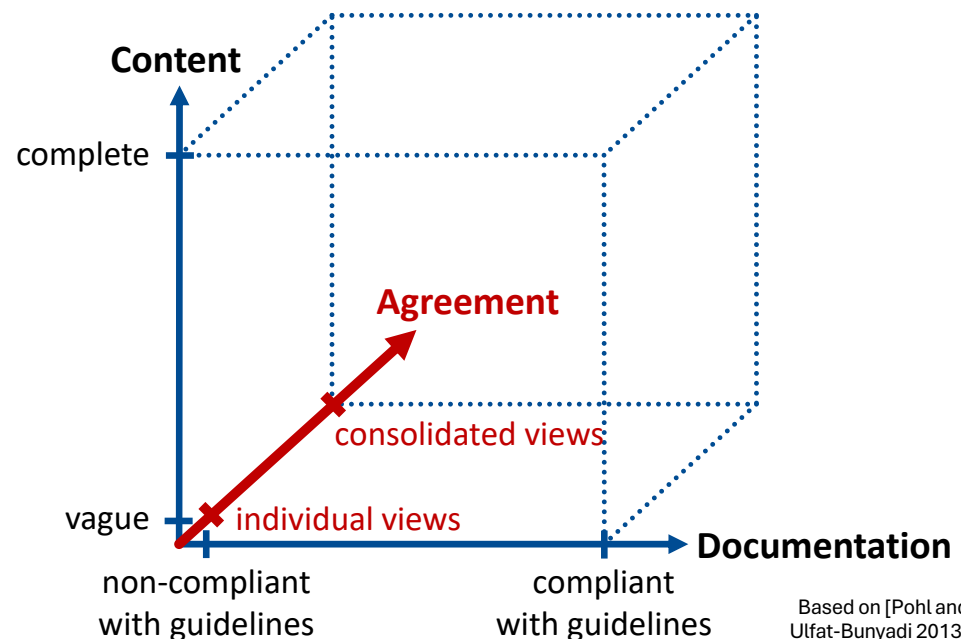
Goal of Requirements Engineering – Three Dimensions of Requirements Engineering

Agreement Dimension

- Focus: Level of agreement achieved between the stakeholders about the requirements and the context; detecting and resolving conflicts
- At the beginning of the requirements engineering process: Stakeholders typically have different, conflicting views about the requirements and the context of the system



At the end:
The stakeholders should have established a sufficient agreement about the requirements and the context



Based on [Pohl and Ulfat-Bunyadi 2013]

Orthogonality of Content and Agreement

- A complete understanding of a requirement or context information does not imply a sufficient agreement between the stakeholders about the requirement or context information
- An agreement between stakeholders about a requirement or context information does not imply a complete understanding of this requirement or context information
- Progress in the agreement dimension can lead to regression in the content dimension, e.g., a creative solution for a conflict leads to new requirements, which might not be well understood
- Progress in the content dimension can lead to regression in the agreement dimension, e.g., new requirements or context information may cause new conflicts

Orthogonality of Documentation and Content

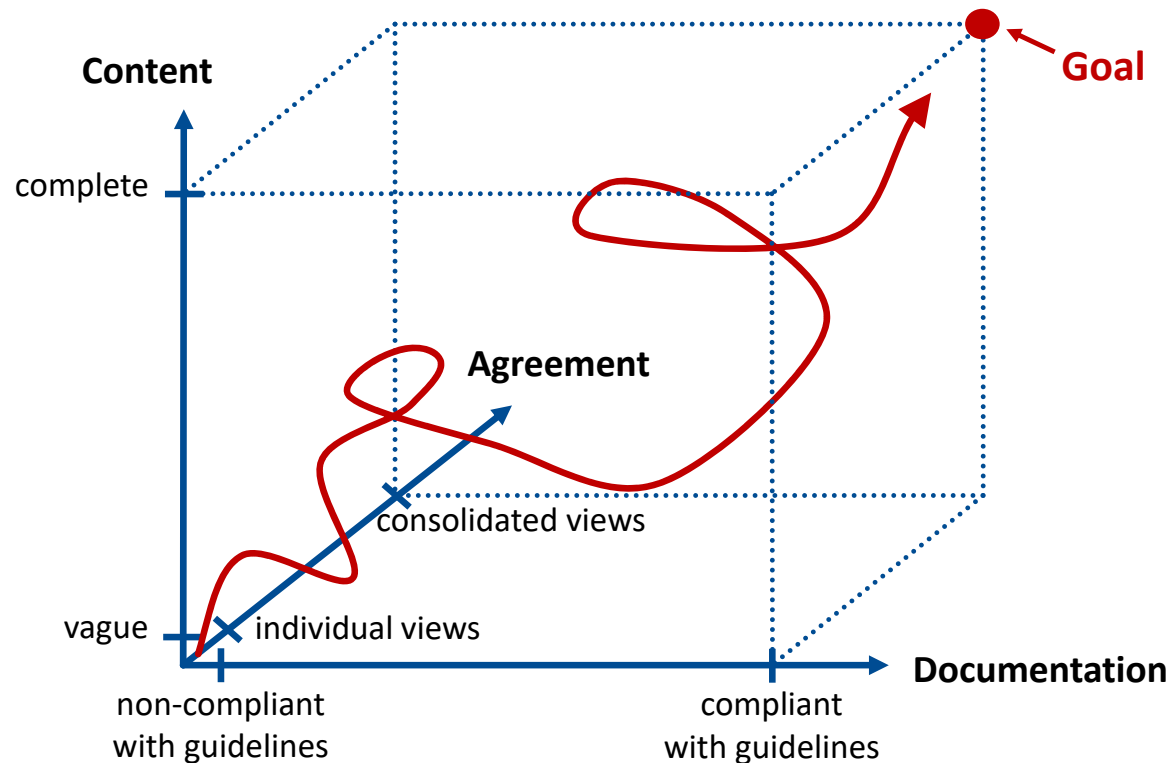
- Compliance of a requirement or context information to the documentation guidelines does not imply a complete understanding of the requirement or context information
- A complete understanding of a requirement or context information does not imply the compliance of the documentation with the documentation guidelines
- Progress in the documentation dimension can lead to regression in the content dimension, e.g., documenting a requirement according to the guidelines can uncover missing information about requirements or the context
- Progress in the content dimension can lead to regression in the documentation dimension, e.g., the documentation of a new elicited requirement does typically not comply with the documentation guidelines

Orthogonality of Agreement and Documentation

- **Agreement** between stakeholders about a requirement or context information does not imply compliance of the documentation with the documentation guidelines
- **Compliance** of a requirement or context information with the documentation guidelines does not imply a sufficient stakeholder agreement
- **Progress** in the documentation dimension can lead to **regression** in the agreement dimension, e.g., documenting a requirement (aligned with the documentation guidelines) can uncover a conflict about the requirement
- **Progress** in the agreement dimension can lead to **regression** in the documentation dimension, e.g., the stakeholders solve a conflict by agreeing on a new requirement which is stated in the minutes (but not documented according to the documentation guidelines)

Goal of Requirements Engineering

Requirements Engineering Process within the Three Dimensions



Based on [Pohl and Ulfat-Bunyadi 2013]

Goal of Requirements Engineering

(Three Goals of) Requirements Engineering

- D** Requirements engineering (RE) is a cooperative, iterative and incremental process which aims at ensuring that:
- (1) All relevant requirements and context information are explicitly known and understood at the required level of detail.
 - (2) A sufficient agreement about the requirements and the context is achieved between the relevant stakeholders.
 - (3) All requirements and context information are documented and specified in compliance with the documentation/specification guidelines.

Overview of the Framework

Motivation for the Framework

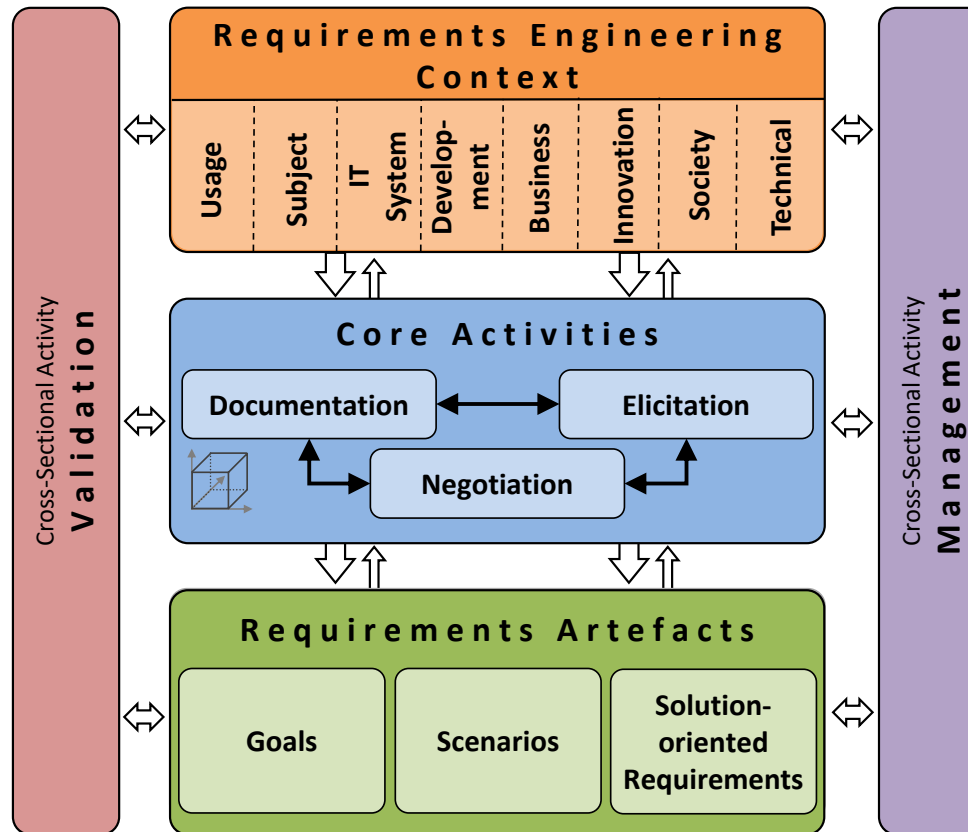
- Structuring requirements engineering by defining a set of core activities and concepts relevant for every requirements engineering process
- Reference structure for teaching requirements engineering
- Reference structure for industry
 - Training for managers, requirements engineers and developers
 - Analysis of strengths and weaknesses of RE processes

...

**Used by many companies,
organisations and universities**

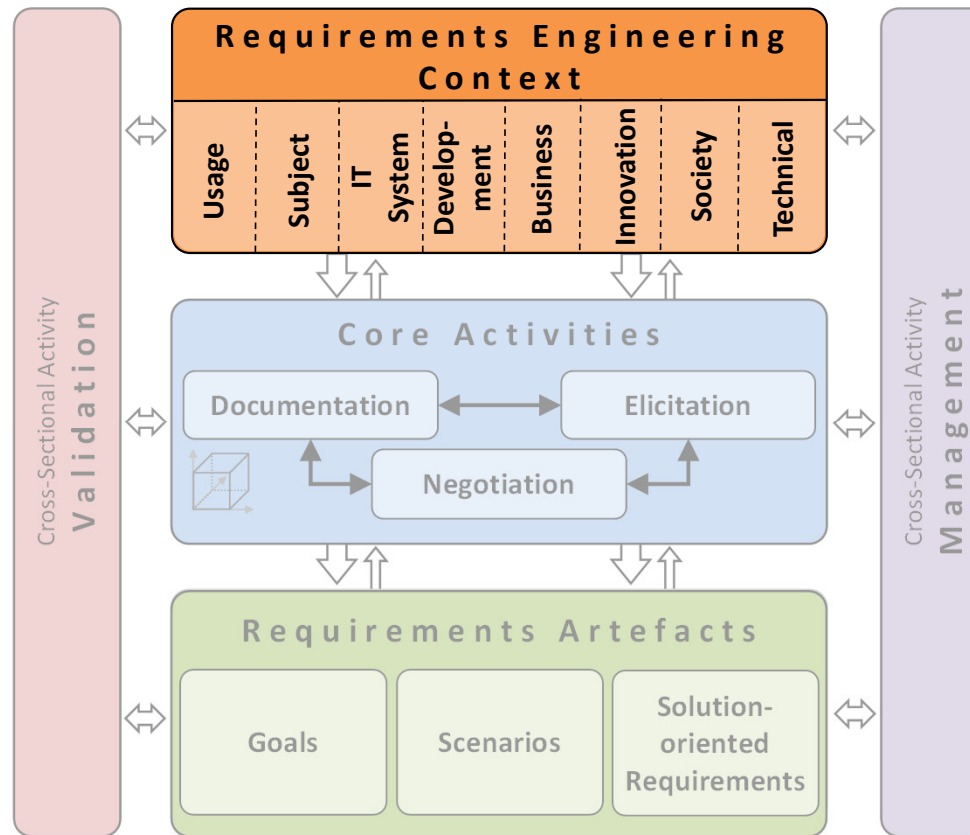
Overview of the Framework

Framework for Requirements Engineering



Requirements Engineering Context

Framework for Requirements Engineering



Context Objects

D A context object is a tangible or intangible object relevant for requirements engineering, which exists in current and/or in future reality.

- Context objects are requirements sources
- Omitting a context object most likely leads to missing and/or wrongly specified requirements

Requirements Engineering Context

Context Information

D Context information is the sub-set of information about the reality relevant for requirements engineering. Context information is typically elicited and – if required – refined during requirements engineering.

- Includes context assumptions:
 - About future situations in which the envisioned system is supposed to be operating (see, e.g., [IREB 2022; Zave and Jackson 1997])
 - About user behaviour, expected workload etc.
 - Can change over time – periodically need validation

Context Objects and Context Information

- Are crucial for all requirements engineering activities as well as for subsequent development activities
- Are identified/elicited from requirements sources (stakeholders, documents, existing systems, data)
- Have to be documented

Requirements Engineering Context

D The requirements engineering context subsumes all context objects and context information (relevant for requirements engineering).

- Context consideration is crucial for requirements engineering as well as other development activities (e.g., design and testing)

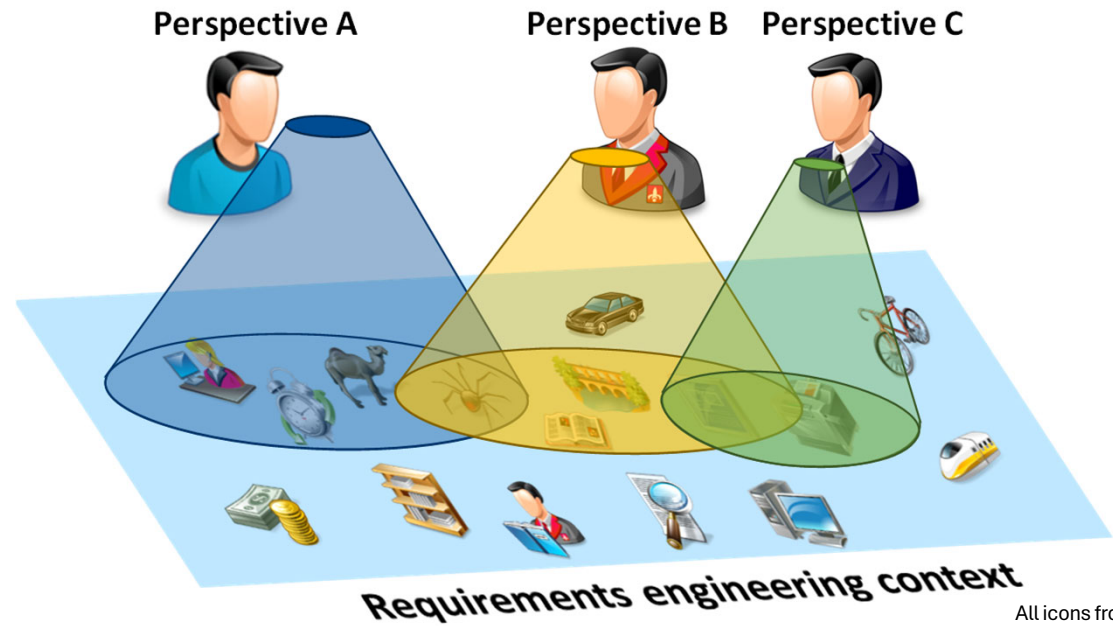
Requirements Engineering Context

Context Perspectives

D A perspective is “a particular way of considering something”.

[Cambridge, 2022]

Perspectives support
handling the complexity
of the requirements
engineering context



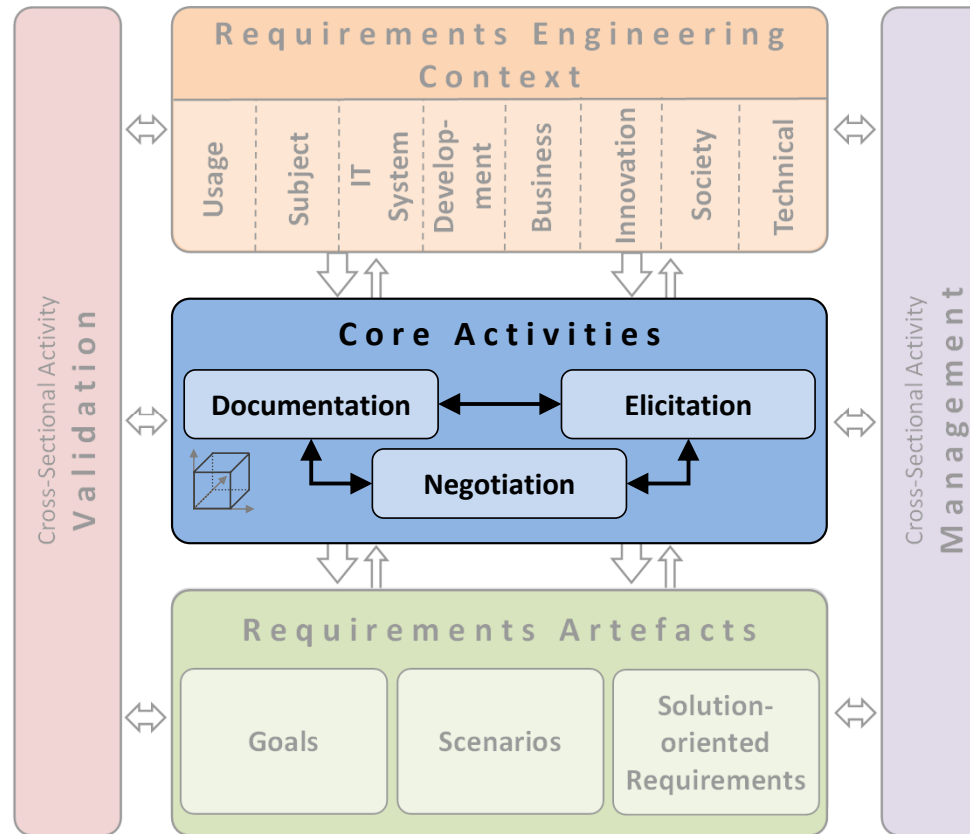
All icons from [1]

Eight Context Perspectives

- Usage perspective
- Subject perspective
- IT system perspective
- Development perspective
- Business perspective
- Society perspective
- Innovation perspective
- Technical perspective

Core Activities

Framework for Requirements Engineering



Core Activities

Elicitation

- D** The goal of the elicitation core activity is to:
- (1) **Identify** and select relevant **requirements sources**
 - (2) Elicit **existing requirements** and context information
 - (3) Develop **new and innovative requirements**
- Requirements sources are **typically not known** at the beginning of the requirements engineering process, and have to be identified through systematic **context consideration**
 - Requirements may **exist** (e.g., stakeholders know them) and can be **elicited from requirements sources**
 - **New** and **innovative requirements** have to be **developed** using techniques that foster **creativity**

Progress in content dimension!

Core Activities

Documentation

- D** The goal of the documentation core activity is to:
- (1) **Document** and **specify requirements** according to the documentation/specification **guidelines** defined for the project
 - (2) **Document context information** according to the documentation **guidelines**
 - (3) **Document process information** according to the documentation **guidelines**

- Early in requirements engineering, information is often **unstructured** and documented **informally**, i.e. **not compliant** with the documentation and specification **guidelines/rules**

Progress in documentation dimension!

Core Activities

Negotiation

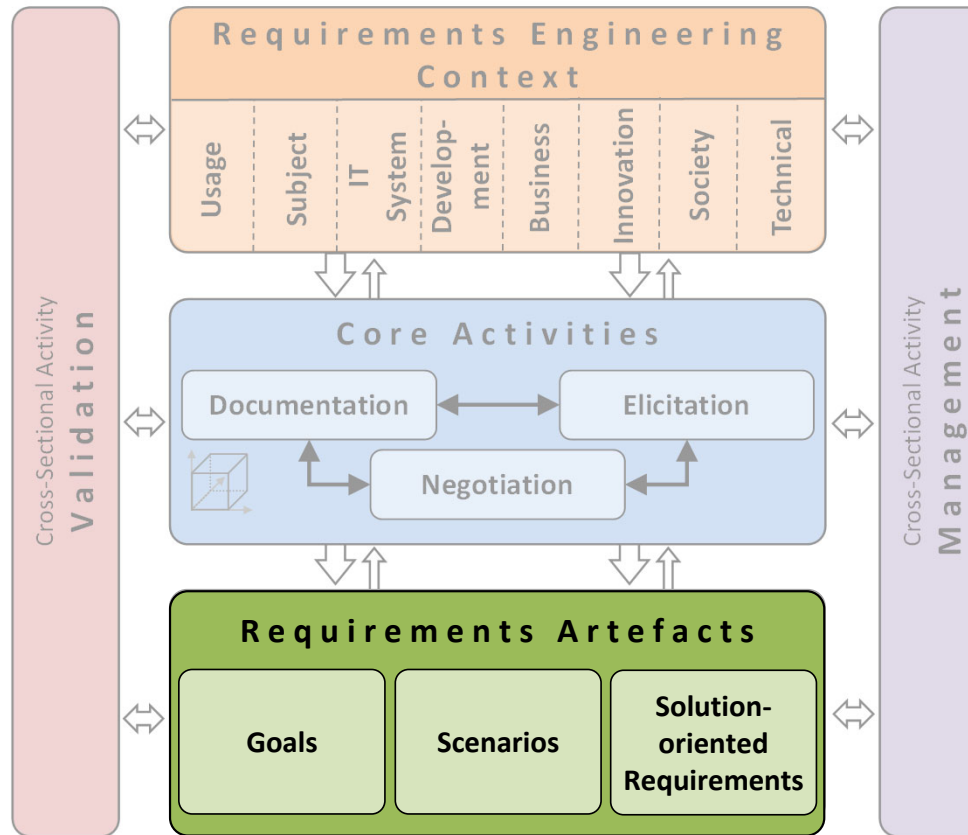
- D** The goal of the negotiation core activity is to:
- (1) Identify conflicts
 - (2) Analyse conflict causes
 - (3) Resolve the conflicts
 - (4) Document rationales and conflict resolutions

- Wishes and needs of the stakeholders, as well as their understanding of the context typically differ and can be in conflict
- Conflicts should be identified and be resolved
- How the conflicts are resolved should be documented

Progress in agreement dimension!

Requirements Artefacts

Framework for Requirements Engineering



Requirements Types vs. Requirements Artefacts

Types of requirements

- Functional requirements
- Quality requirements
- Constraints

Requirements artefacts

- Goals
- Scenarios
- Solution-oriented requirements

A requirements artefact (goal, scenarios and solution-oriented requirement) can be used to document functional and quality requirements as well as constraints

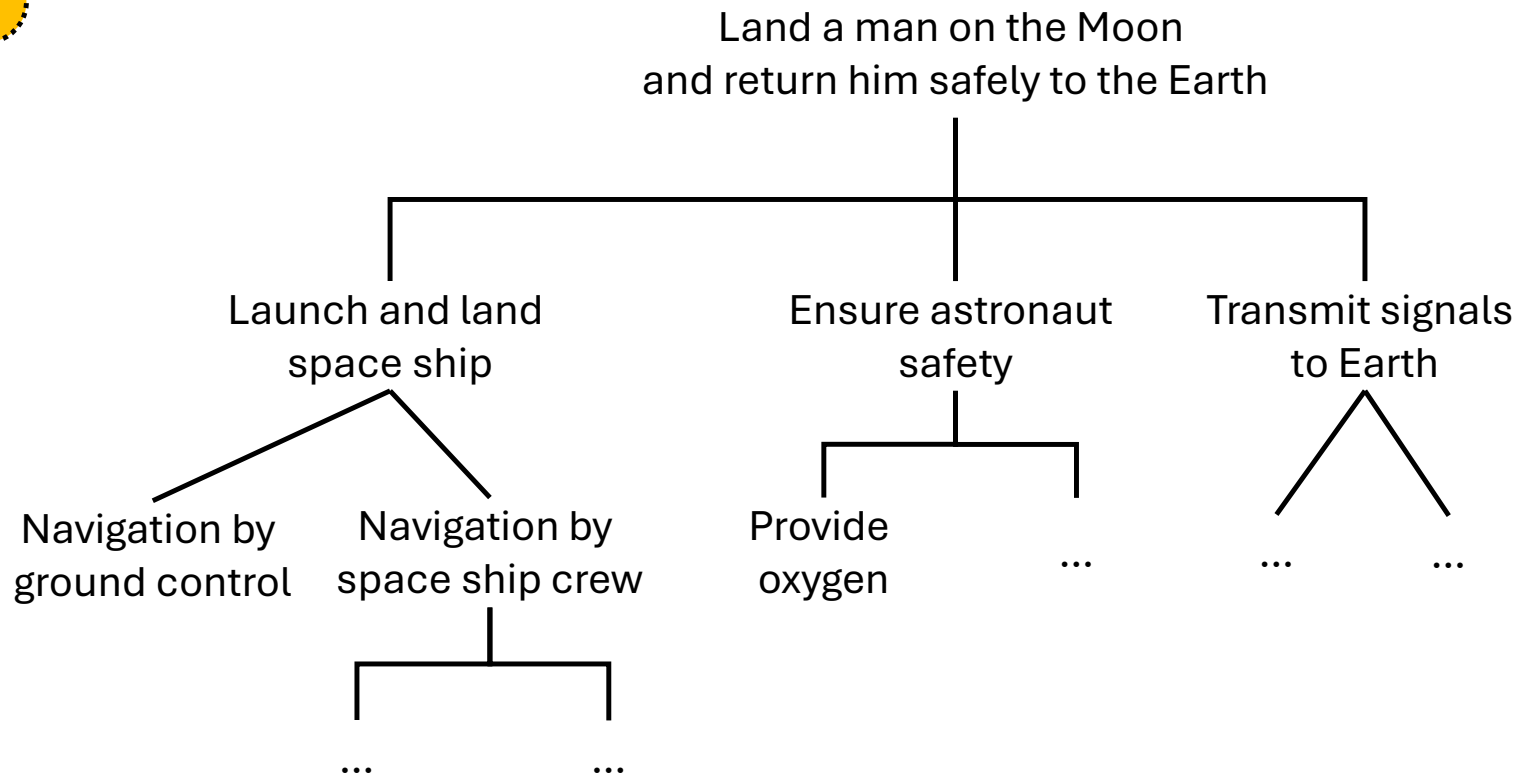
Goals (1)

D A goal describes an intention of one or more stakeholders with regard to the objectives, properties or use of the system.

Characteristics of a goal:

- Refines the system vision into more detailed objectives to be fulfilled by the system
- Should be solution free

Goals (2)



Scenarios (1)

D A scenario describes a concrete example of satisfying or failing to satisfy a goal (or set of goals). A scenario often defines a sequence of interaction steps between the system and its users (humans and/or other systems).

Characteristics of a scenario:

- Typically documents a concrete example of system usage
- Increases the comprehensibility of goals
- Puts requirements into context

Requirements Artefacts

Scenarios (2)



Scenario “Withdraw money”:

Chris inserts her bank card into the slot of the ATM (automated teller machine). Chris enters her personal identification number and the amount to withdraw. The ATM informs Chris that withdrawing the desired amount is not possible because the amount exceeds her balance.

Requirements Artefacts

Scenarios (2)



Scenario “Automatic braking manoeuvre”:

As the distance between the two cars decreases rapidly, there is a high risk of a rear-end collision. A rapid change of lane might cause the car to skid or spin since the car is driving on the motorway at a high speed of more than 55 mph. Hence, prior to changing lane, the speed of the car must be reduced. Consequently, the on-board computer initiates an automatic emergency braking manoeuvre. As the antilock braking system ensures that the car remains steerable during a braking manoeuvre, Carl could safely change lane during the manoeuvre. In order to avoid the driver being startled by the automatic braking, the system informs Carl about the initiation of the emergency braking. After having restored a safe distance to the car driving ahead, the on-board computer passes control back to Carl. The system notifies Carl about the passing over of control in order to allow him to prepare to take over control.

Solution-oriented Requirements

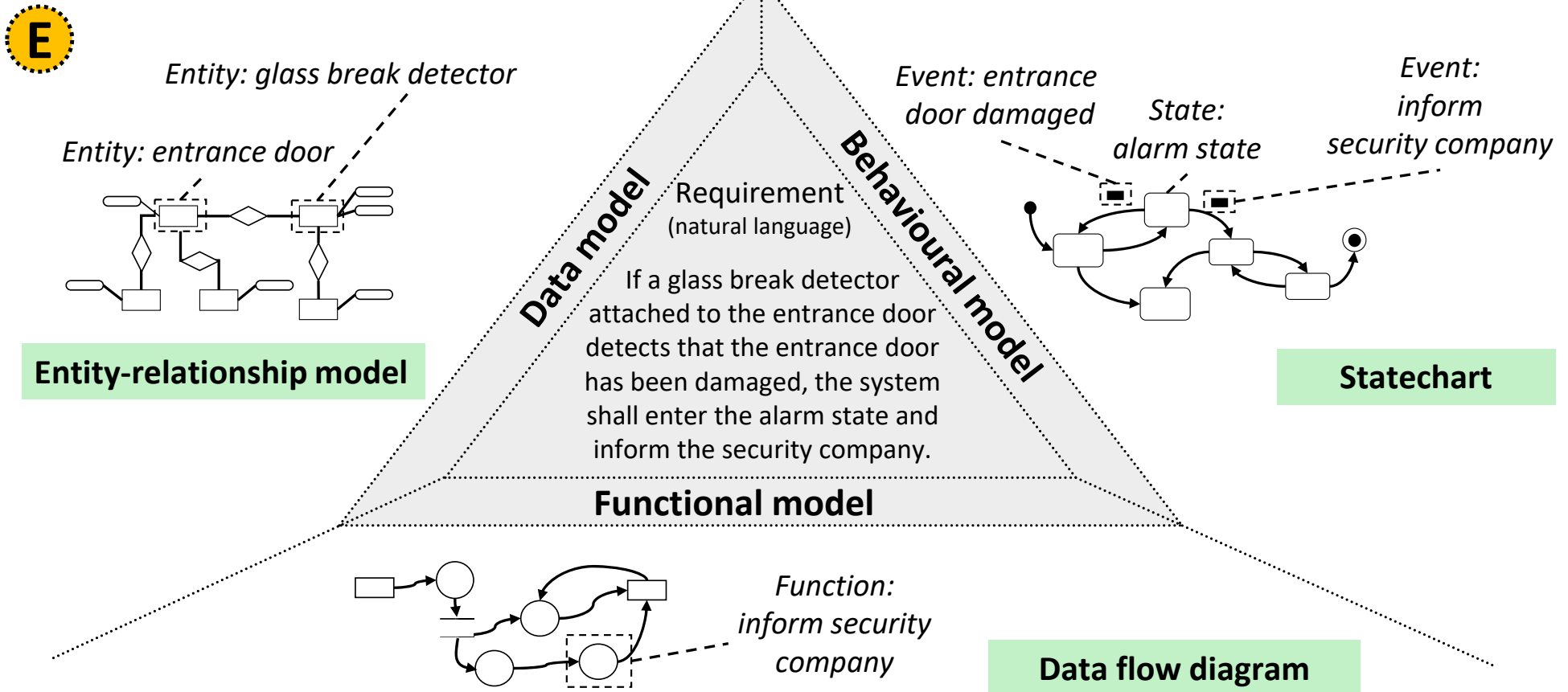
Characteristics of solution-oriented requirements:

- **Specify** requirements at a level of detail **sufficient for later development activities** such as design and test
- Often imply a **conceptual/logical solution**
- Should be **conflict-free**
- Should be **agreed** on by all (relevant) stakeholders
- Should be as **complete** as possible

Three Types of Solution-oriented Requirements (1)

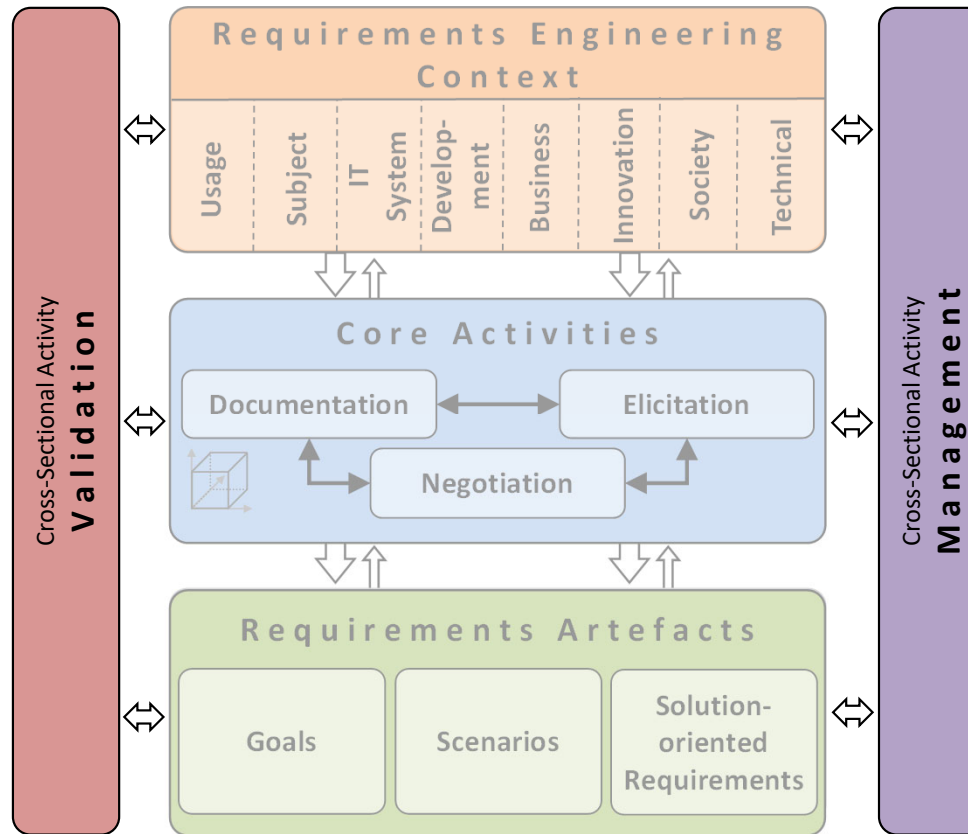
- (1) **Data perspective**: Definition of the static data structures, including data types, attributes and relationships
- (2) **Functional perspective**: Definition of the systems functions, including the data processing and transformation of system inputs into system outputs
- (3) **Behavioural perspective**: Definition of the system behaviour, including reactions to external stimuli in form of permitted states, transitions and outputs

Three Types of Solution-oriented Requirements (2)



Cross-Sectional Activities

Framework for Requirements Engineering



Cross-Sectional Activities

Validation Goals

- (1) Validation of the context consideration, including checking whether the eight context perspectives have been considered adequately
- (2) Validation of the execution of requirements engineering activities, including checking if the activities have been performed in compliance with process definitions
- (3) Validation of requirement artefacts and context information in terms of content, documentation and agreement achieved

Management Goals

- (1) Management of the requirements engineering context including identification of context changes, change impact analysis, and context consideration in activities
- (2) Management of requirements engineering activities, including planning and controlling the execution of activities
- (3) Management of the requirements artefacts and context information, including persistent recording, prioritisation, traceability, configuration and change management

Summary

- Requirements engineering: Establishing a vision in context
- Vision guides context consideration and justifies expenses
- Three dimensions of requirements engineering: Agreement, Content, Documentation
- Goal of requirements engineering:
 - Sufficient agreement established
 - Requirements and context information documented according to the guidelines
 - Requirements and context sufficiently understood (content)
- Framework for requirements engineering (four building blocks):
 - Requirements engineering context: structured into eight perspectives
 - Three core activities: documentation, elicitation and negotiation (three dimensions)
 - Three types of requirements artefacts: Goals, scenarios and solution-oriented requirements
 - Two cross-sectional activities: Validation and management

Literature References

- [Cambridge 2022] Cambridge Dictionary: English Meaning of Perspective.
<https://dictionary.cambridge.org/dictionary/english/perspective> accessed on 15/05/2023.
- [Dudley 2000] B. Dudley (Ed.): The Greatest Speeches of President John F. Kennedy. Titan, West Vancouver, 2000.
- [IREB 2022] IREB Certified Professional for Requirements Engineering, Foundation Level, Syllabus, Version 3.1.0, September 2022, https://www.ireb.org/content/downloads/2-cpre-foundation-level-syllabus-3-0/cpre_foundationlevel_syllabus_en_v.3.1.pdf; accessed on 15/05/2023.
- [Pohl and Ulfat-Bunyadi 2013] K. Pohl, N. Ulfat-Bunyadi: The Three Dimensions of Requirements Engineering: 20 Years Later. In J. Bubenko et al., eds. Seminal Contributions to Information Systems Engineering. Springer Berlin Heidelberg, pp. 81–87, 2013.
- [Zave and Jackson 1997] P. Zave, M. Jackson: Four Dark Corners of Requirements Engineering. ACM Transactions on Software Engineering and Methodology, Vol. 6, No. 1, 1997, pp. 1-30.

Image References

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Legend

 Definition

 Example